

Project Design Phase-II

Technology Stack (Architecture & Stack)

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Project Name:	Automated Network Request Management System
Maximum Marks:	4 Marks

Technical Architecture Overview

The system is designed as an enterprise-grade, cloud-native application. It bridges the gap between front-end end-user catalog submissions and direct backend hardware asset table persistence through an automated low-code workflow orchestration model.

Structural Breakdown:

- **Frontend Tier (Local Client Interaction):** Users interact with a clean Service Portal built using modern web standards. Client-side processing handles UI dynamic logic, instant context retrieval, and validation filters.
- **Process Orchestration Engine (Cloud Logic):** Triggers server-side logic pathways via ServiceNow Flow Designer and Workflow Studio. This engine manages operational gates, rules processing, and line-manager approval loops.
- **Database Ledger Store (Persistence Layer):** Securely seals validated data inputs directly into custom enterprise database schemas without manual text copy-pasting.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Customer-facing Web portal interface where users search, select request options, and view live stage progress badges.	ServiceNow Service Portal, HTML5, CSS3, JavaScript (ES6), Bootstrap Framework.

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2.	Application Logic-1	Client-side validation scripts, session properties query rules, and dynamic field visibility behaviors.	ServiceNow Client Scripts, Catalog UI Policies, GlideUser (g_user) API.
3.	Application Logic-2	Server-side background execution workflows that intercept user checkout paths and check rule conditions.	ServiceNow Flow Designer, Workflow Studio Actions, Business Rules.
4.	Application Logic-3	Cross-table routing calculations and relational approval mapping generation handlers.	ServiceNow GlideRecord API, JavaScript Engine.
5.	Database	Local system data dictionary structures holding system transactional variables and workflow runtime metrics.	ServiceNow Database Layer (built on MySQL/MariaDB foundation core).
6.	Cloud Database	Permanent, structural production relational table hosting the master infrastructure asset configurations.	Custom System Schema Table (u_network_database).
7.	File Storage	Secure storage for transactional audit history logs and physical office floor plan document attachments.	ServiceNow Managed Cloud Attachment Storage (sys_attachment).
8.	External API-1	Identity contextual information query engine mapping incoming user sessions to corporate organizational directories.	ServiceNow internal GlideSystem session context lookup API.

S.No	Component	Description	Technology
9.	External API-2	Automated outbound notifications providing line managers with one-click decision-making functions.	ServiceNow Cloud Mail Server Engine (SMTP/IMAP Notification Engine).
10.	Machine Learning Model	Optional optimization rule set used to automatically assign fulfiller tasks based on agent workload queues.	ServiceNow Predictive Intelligence Engine (Optional Similarity/Clustering Framework).
11.	Infrastructure (Server / Cloud)	High-availability cloud host layer ensuring multi-tenant isolation, data separation, and secure runtime boundaries.	ServiceNow Enterprise Cloud Platform-as-a-Service (PaaS) Instance.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Integration of universally supported front-end grid layouts to ensure responsive UI display across various client browser viewports.	Bootstrap CSS Framework, AngularJS (ServiceNow Core UI framework stack wrapper).
2.	Security Implementations	Data encryption definitions safeguarding active sessions, access control parameters, role profiles, and platform threat protection rules.	Advanced Encryption Standard (AES-256), Multi-Factor Authentication (MFA), Access Control Lists (ACLs), ServiceNow role-based access management.
3.	Scalable Architecture	Logical layer breakdown strategy that isolates user presentation elements	Multi-tier Cloud Platform-as-a-Service (PaaS) framework,

S.No	Characteristics	Description	Technology
		from backend ledger transactions to manage fluctuating query traffic.	microservices-inspired multi-node cloud application nodes.
4.	Availability	Resilient architectural planning guaranteeing zero service interruptions via continuous redundancy paths and fallback cluster instances.	ServiceNow Advanced High Availability (AHA) Architecture, geo-redundant automated data center replication.
5.	Performance	Computational tuning to enforce minimal form load delays, sub-second execution speeds, and background workflow balancing processing.	ServiceNow Client-Side Cache Optimization, Indexed Table fields on u_network_database, Asynchronous Background Workers.